

Dataset / Column	Column Description	Data Type	Valid Values	Error Handling Rules	Missing Value Handling Rules	Used in Analysis	Notes	FAQ	
Spend	Marketing spend and campaign activity data: impressions, clicks and spend recorded by source, campaign, ad group and ad.								
Date	Spend date.	Date	Valid calendar date within the analysis period.	If the date has an invalid format or falls outside the analysis period, review the row separately. If the date cannot be restored, exclude the row from time-series analysis.	Missing values are not allowed. Rows with missing dates are excluded from analysis.	Yes	Key field for analyzing marketing spend dynamics.	-	
Source	Paid acquisition channel.	Categorical string	Bloggers; CRM; Facebook Ads; Google Ads; Offline; Organic; Partnership; Radio; SMM; Telegram posts; Test; Tiktok Ads; Webinar; Youtube Ads.	If the value has a different spelling but the source is clear, standardize it. If the source is unknown or cannot be interpreted, mark it as Unknown or exclude it from source-level analysis.	Missing values are not allowed for marketing analysis. Rows with missing source values should be reviewed; if the source cannot be restored, mark as Unknown or exclude from channel analysis.	Yes	Key field for linking marketing spend with leads/deals and analyzing channel efficiency.	-	
Campaign	Campaign where the advertising budget was spent.	String / category	Campaign names from the export.	If one campaign has multiple spellings, standardize them. If the value is technical noise and cannot be interpreted, mark it as Unknown. Dates inside campaign names should not be treated as actual campaign launch dates.	Missing values are acceptable when Spend = 0. If Spend <> 0, fill the field using the rule: Date + "_" + Source.	Yes	Key field for linking with Deals. Campaign names may be non-standardized. Used for campaign efficiency analysis and linking campaigns to deals.	Campaign is an internal campaign name defined by marketing. There are no strict naming rules. Dates inside campaign names should not be used as actual launch dates. Leads are linked to this field, showing which campaign generated the lead.	
Impressions	Number of ad impressions on a given date.	Integer, non-negative (>= 0)	0 and above.	Negative values should be treated as errors and excluded from calculations. Non-numeric values should be converted to numeric where possible; otherwise treated as errors and excluded.	Missing values are not allowed. If the value is missing and cannot be restored, set it to 0.	Yes	Used for CTR calculation and reach analysis.	-	
Spend	Advertising spend in euros.	Float, 2 decimals, non-negative (>= 0)	0, 0.01 and above.	Negative values should be treated as errors and excluded from calculations. Non-numeric values should be converted to numeric where possible; otherwise excluded from marketing spend calculations.	Missing values are not allowed. If spend is missing, set it to 0.	Yes	Key field for unit economics and marketing efficiency analysis.		
Clicks	Number of unique clicks on an advertising link on a given date.	Integer, non-negative (>= 0)	0 and above.	Negative values should be treated as errors and excluded from calculations. If Clicks > Impressions, review the row as a logical error.	Missing values are not allowed. If the value is missing and cannot be restored, set it to 0.	Yes	Used for CTR, CPC and traffic quality analysis.		
AdGroup	Ad group name.	String / category	Ad group names from the export.	If one ad group has multiple spellings, standardize them. If the value cannot be interpreted, mark it as Unknown.	Missing values are acceptable if the ad group has no ad-level detail. If the group is detailed by Ad, create a new group using the Ad name.	Optional	Can be linked to the Term field in Deals. Can be used for deeper advertising analysis, but is not required for the base project.		
Ad	Advertisement name.	String / category	Ad names from the export.	If one ad has multiple spellings, standardize them. If the value cannot be interpreted, mark it as Unknown.	Missing values are acceptable.	Optional	Can be linked to the Content field in Deals. Can be used for creative analysis, but is not required for the base project.		
Contacts									
Id	Contact ID.	String	Unique non-empty contact identifier.	If the value is not unique, review the row separately. If rows are full duplicates, remove them. If other fields differ, select the latest/most relevant record or mark as a conflict. If the ID cannot be restored, mark as Unknown and do not use in calculations or table joins.	Missing values are not allowed. Rows without Id are marked as Unknown and excluded from calculations and joins.	Yes	Key field for linking contacts with deals and calls.		
Contact Owner Name	Sales manager responsible for the contact.	String / category	Manager name in CRM format: First Name Last Name.	If one manager has multiple spellings, standardize them. If the value cannot be interpreted, mark it as Unknown. For interpretation, use first-contact information from the Calls dataset.	Missing values are undesirable. If the contact owner is missing, mark as Unknown, but do not remove the entire row.	Optional	Can be used for additional workload/sales activity analysis, but the main sales analysis is based on Deal Owner Name.		
Created Time	Contact creation timestamp.	Date / time	Valid contact creation date and time.	If the date is invalid, try to convert it to datetime. If it cannot be restored, exclude the row from contact time-series analysis.	Missing values are not allowed for contact time-series analysis. If the date is missing, do not use the row for analyzing contact creation dynamics.	Optional	Can be used for analyzing contact creation dynamics.		
Modified Time	Timestamp of the latest contact update.	Date / time	Valid contact modification date and time.	If the date is invalid, try to convert it to datetime. If Modified Time is earlier than Created Time, review the row as a logical error.	Missing values are undesirable but not critical. If the field is missing, keep the row, but do not use it for contact update analysis.	Optional	Can be used to check contact updates, but is not a key field for the main analysis.		
Deals									
Main deals table: deal status, responsible manager, lead source, campaign, lead quality, product, payment type, amounts, client city, German language level and creation/closing dates.									
Business rules: (1) Successful deal = Stage == Payment Done. (2) Revenue is calculated from Initial Amount Paid only for successful deals. (3) If Stage == Payment Done but Initial Amount Paid is missing, the deal is included in conversion but excluded from revenue. (4) Rows with Lost Reason == Duplicate are excluded from loss and conversion analysis after validation, as they may represent technical duplicate leads.									
Id	Deal ID.	String	Unique non-empty deal identifier.	If the value is not unique, review the row separately. Full duplicates should be removed after validation. If the ID cannot be restored, the row cannot be used as a unique deal and should be excluded from analysis.	Missing values are not allowed. Rows without Id are excluded from deal analysis.	Yes	Technical deal key; should be converted to string.		
Deal Owner Name	Sales manager responsible for the client deal. One client may have several deals.	String / category	Manager name in CRM format: First Name Last Name.	If one manager has multiple spellings, standardize them. If the value cannot be interpreted, mark it as Unknown.	Missing values are undesirable. If the manager is missing, keep the row in the general analysis, but exclude it from manager-level analysis.	Yes	Key field for sales team performance analysis.		
Closing Date	Closing date; for successful deals, this is the payment date.	Date	Valid calendar date.	If the date cannot be converted to datetime, mark it as an error. If Closing Date is earlier than Created Time, review it as a logical error. For successful deals, the closing date should reflect the payment date.	Missing values are acceptable for open or unfinished deals. For deal closing time analysis, only rows with a completed date are used.	Yes	Used for deal duration analysis and time-series analysis.		
Quality	Deal quality classification indicating potential or target status. Filled subjectively by sales managers using an ABCDE scale, where A is a hot lead and E is a non-qualified lead with incorrect data or no contact.	Categorical string	A - High; B - Medium; C - Low; D - Non Target; E - Non Qualified; F.	Standardize different spellings of the same category. Mark unknown values as Unknown.	Missing values are acceptable, but should be separated into the Unknown group for lead quality analysis.	Yes	Important field for lead quality analysis by source. Filled subjectively by sales managers.	A deal is closed when the manager stops communication with the client. For students, this is the payment date. If the closing date is earlier than creation date, this is an error and should be handled with an explained analytical decision.	
Stage	Current deal stage. Payment Done means the deal has been paid.	Categorical string	CRM statuses, for example Lost, Payment Done and other funnel stages.	Standardize different spellings of the same status. Mark unknown values as Unknown. Empty status is an error for funnel analysis.	Missing values are not allowed for funnel analysis. If the status cannot be restored, exclude the row from status/conversion analysis.	Yes	Main field for defining the funnel and deal success.	Payment Done means that money was received. Other statuses are not considered payments; they can be translated and interpreted for funnel analysis.	
Lost Reason	Reason why the client declined the purchase and moved to Lost.	Categorical string	CRM loss reasons, for example Doesn't Answer, Duplicate and others.	Standardize different spellings of the same reason. Use Duplicate to validate duplicates and exclude them from part of the analysis.	Missing values are acceptable for deals that are not in Lost. For lost deals, a missing reason means incomplete information and should be marked as Unknown.	Yes	Used for analyzing reasons for lost leads.	Duplicate means there were two leads and one of them is marked as Lost because it is a duplicate. Waiting for payment or dropped statuses are not payments; payment fact is determined by Stage.	
Page	Website page where the lead submitted the request.	String	URL or page path from CRM, e.g. /eng.	Standardize different spellings of the same page if needed for analysis. Mark technical noise as Unknown.	Missing values are undesirable but not critical. If the page is missing, keep the row in the general analysis and mark it as Not page.	Optional	Can be used for additional landing page analysis.		
Campaign	Marketing campaign that generated the lead.	String / category	Campaign names from the export.	If one campaign has multiple spellings, standardize them. If the value is technical noise and cannot be interpreted, mark it as Unknown. Dates inside campaign names should not be treated as actual campaign launch dates.	Missing values are acceptable if the lead did not come from an advertising campaign. For campaign analysis, rows with missing campaign are not used.	Yes	Used for linking with Spend and analyzing campaign effectiveness.		
SLA	Sales response time: time between the lead submission and the first sales contact.	Time	Time format, e.g. 00:10:11; non-negative value.	Convert invalid formats to a time format where possible. Negative or non-interpretable values should be marked as errors/Unknown and excluded from SLA analysis.	Missing values are acceptable, but rows without SLA are not used in sales response speed analysis.	Yes	Important field for the hypothesis about response speed and conversion.		
Content	Ad that generated the lead.	String / category	Ad names from CRM, matching Spend.Ad.	Standardize different spellings of the same creative if the field is used. Mark technical noise as Unknown.	Missing values are acceptable. For creative analysis, rows with missing values are not used.	Optional	Can be linked to Spend.Ad, but is not required for the base analysis.		
Term	Ad group that generated the lead.	String / category	Ad group names from CRM, matching Spend.AdGroup.	Standardize different spellings if the field is used. Mark technical noise as Unknown.	Missing values are acceptable. For AdGroup/Term analysis, rows with missing values are not used.	Optional	Can be linked to Spend.AdGroup, but is not required for the base analysis.		
Source	Marketing channel that generated the lead.	Categorical string	Bloggers; CRM; Facebook Ads; Google Ads; Offline; Organic; Partnership; Radio; SMM; Telegram posts; Test; Tiktok Ads; Webinar; Youtube Ads.	If the value has a different spelling but the source is clear, standardize it. If the source is unknown or cannot be interpreted, mark it as Unknown or exclude it from source-level analysis.	Missing values are not allowed for source analysis. Rows with missing source values should be reviewed; if the source cannot be restored, mark as Unknown or exclude from channel analysis.	Yes	Key field for marketing source analysis.		
Payment Type	Client payment method.	Categorical string	CRM payment types, e.g. Recurring Payments and other export values.	Standardize different spellings of the same payment type. Mark unknown values as Unknown.	Missing values are acceptable for unpaid deals. For payment type analysis, only rows with filled values are used.	Yes	Filled only for a limited part of deals; use carefully.		
Product	Product purchased by the client or considered for purchase.	Categorical string	Data Analytics; Digital Marketing; Find yourself in IT; UX/UI Design; Web Developer.	Standardize different spellings of the same product. Mark unknown values as Unknown.	Missing values are acceptable for early-stage or unsuccessful deals. For product analysis, rows without a product are excluded or separated as Unknown.	Yes	Key field for product analysis and unit economics.		
Education Type	Learning format suitable for the client.	Categorical string	Evening; Morning.	Standardize different spellings of the same format. Mark unknown values as Unknown.	Missing values are acceptable for deals that did not reach format selection. For learning format analysis, rows without a value are not used.	Yes	Used for learning format analysis.		
Created Time	Deal creation timestamp.	Date / time	Valid deal creation date and time.	If the value cannot be converted to datetime, mark it as an error. Dates outside the analysis period should be reviewed separately.	Missing values are not allowed for time-series analysis. If the date cannot be restored, exclude the row from deal creation dynamics analysis.	Yes	Key field for time-series analysis and deal duration calculation.		
Course duration	Course duration in months.	Integer, non-negative (>= 0)	Value greater than 0, e.g. 6 or 11 months.	Negative, zero or non-numeric values should be reviewed. If the value cannot be interpreted, do not use it for course duration analysis.	Missing values are acceptable if no product/course was selected. Rows with missing values are not used for course duration analysis.	Yes	Can be used in product analysis.		
Months of study	Number of months the student has studied by the time the dataset was created. It does not show whether the student is still studying.	Integer, non-negative (>= 0)	Value from 0 to Course duration.	Negative values should be treated as errors. Values greater than Course duration should be reviewed as logical errors.	Missing values are acceptable if the client did not become a student. For student analysis, only filled values are used.	Yes	Can be used as a proxy for retention / study duration.	If this field is empty, the student is not studying and their money should not be included, regardless of whether this is a data error or a refund case.	
Initial Amount Paid	Amount paid by the client as the first payment. It may be the full tuition fee or only part of it.	Float, 2 decimals, non-negative (>= 0)	Value >= 0.	Convert non-numeric values to numeric where possible. Negative values should be treated as errors. If Initial Amount Paid > Offer Total Amount, review it as a logical error.	Missing values are acceptable for unpaid deals. For Payment Done deals, missing amount means payment without a recorded amount; include such deals in conversion, but not in revenue.	Yes	Key field for revenue calculation.	If Stage is Payment Done, this amount is treated as paid. Payment fact is determined by a combination of fields, but Stage is the primary business rule. Values 0, 1 and 9 may represent demo access or symbolic payments.	
Offer Total Amount	Full contract tuition value: how much the client will pay if they complete the course and make all payments.	Float, 2 decimals, non-negative (>= 0)	Value >= 0.	Convert non-numeric values to numeric where possible. Negative values should be treated as errors. If the value is lower than Initial Amount Paid, review it as a logical error.	Missing values are acceptable if no offer was created. Use only filled values for average check calculation.	Yes	Used for analyzing potential and actual deal value.		
Contact Name	Contact ID from the Contacts table.	String	Unique contact identifier from Contacts.	If the value is missing or not found in Contacts, review the row. Do not use such rows for joins with Contacts.	Missing values are undesirable. The deal can remain in general analysis but should not be used for joins with Contacts/Calls.	Yes	Despite the field name, it semantically represents contact ID.		
City	Client city.	String / category	City name.	Convert '-', empty strings, technical noise and spelling variants of the same city to a standardized value or Unknown.	Missing values are acceptable. For geographic analysis, rows without city are excluded or grouped as Unknown.	Yes	Used for geographic analysis.		
Level of Deutsch	Client's German language level.	Categorical string	A1, A2, B1, B2, C1, C2 and similar CRM values.	Standardize variants to the language level format. Incorrect or free-text values should be marked as Unknown.	Missing values are acceptable. For German language level analysis, only filled/standardized values are used.	Yes	Filled only for part of clients; use carefully. Requires standardization.		
Calls									
Id	Call ID.	String	Unique non-empty call identifier.	If the value is missing or not unique, review it separately. If rows are full duplicates, remove them. If the identifier cannot be restored, exclude the row from unique call analysis.	Missing values are not allowed. Rows without Id are excluded from call analysis.	Yes	Technical call key; should be treated as an identifier, not as a number.		
Call Start Time	Call start timestamp.	Date / time	Valid call date and time.	If the value has an invalid format, try to convert it to datetime. If the date cannot be restored, do not use the row for call time-series analysis.	Missing values are not allowed. Rows without call start time are not used for call dynamics analysis.	Yes	Key field for analyzing sales activity over time.		
Call Owner Name	Manager who participated in the call.	String / category	Manager name in CRM format: First Name Last Name.	If one manager has multiple spellings, standardize them. If the value cannot be interpreted, mark it as Unknown.	Missing values are undesirable. If the manager is missing, keep the row for total call counts, but exclude it from manager-level analysis.	Yes	Used for analyzing sales team activity.		
CONTACTID	ID of the contact associated with the call.	String	Identifier of a contact from the Contacts table.	If the value is not found in Contacts, review the row. Do not use such rows for joins with contacts.	Missing values are acceptable for general call analysis, but rows without CONTACTID cannot be used for joins with contacts and deals.	Yes	Key field for linking calls with contacts.		
Call Type	Call type: Inbound = received incoming call, Missed = missed incoming call, Outbound = outgoing call.	Categorical string	Inbound; Missed; Outbound.	Standardize different spellings of the same call type. Mark unknown values as Unknown.	Missing values are undesirable. If call type is missing, keep the row but do not use it for call type analysis.	Yes	In the initial audit, most calls are Outbound.		
Call Duration (in seconds)	Call duration in seconds.	Integer, non-negative (>= 0)	0 and above.	Negative values should be treated as errors and excluded from duration analysis. Non-numeric values should be converted to numeric where possible; otherwise exclude the row from duration analysis.	Missing values are undesirable. If duration is missing, set it to 0.	Yes	Important field for analyzing contact quality with the client.		
Call Status	Detailed call status with multiple possible values.	Categorical string	CRM statuses, e.g. Attended, Dialed.	Standardize different spellings of the same status. Mark unknown values as Unknown.	Missing values are undesirable. If the status is missing, keep the row for total call counts, but do not use it for status analysis.	Yes	Important field for analyzing contact success.		
Dialled Number	Client phone number.	String	Phone number if filled.	Because the field has 100% missing values, exclude it from analysis. Do not attempt to restore phone numbers.	The field is completely empty and should be excluded from analysis.	No	The field does not provide analytical value in the current export.		
Outgoing Call Status	Detailed outgoing call status with multiple possible values.	Categorical string	Cancelled; Completed; Overdue; Scheduled.	Standardize different spellings of the same status. Mark unknown values as Unknown.	Missing values are acceptable for non-outbound calls. For outgoing call analysis, rows with missing status are not used.	Optional	Useful for additional outgoing call detail analysis.		
Scheduled in CRM	Attribute indicating whether the call was scheduled in CRM in advance.	Boolean / binary	FALSE; TRUE.	Values other than 0/1 or TRUE/FALSE should be reviewed and converted to binary if the meaning is clear.	Missing values can be interpreted as no information. For scheduled call analysis, rows with missing values are not used.	Optional	Can be used for additional analysis of CRM discipline.		
Tag	Tag that may have been assigned to the call by a manager.	String	CRM tag if filled.	The field is completely empty and should be excluded from analysis.	The field is completely empty and should be excluded from analysis.	No	The field does not provide analytical value in the current export.		

Data Dictionary and Dataset Description				
Purpose	This workbook documents the structure, business meaning, data types, validation rules, missing-value handling rules and analytical usage of the source datasets.			
Datasets	Spend, Contacts, Deals, Calls			
Business rule	A paid deal is defined as Stage = Payment Done.			
Revenue rule	Actual revenue is calculated from Initial Amount Paid only for paid deals.			
Important limitation	Some fields have limited completeness and should be interpreted carefully, especially Payment Type, City and Level of Deutsch.			
Usage	This file supports data audit, cleaning decisions, business rules and portfolio documentation.			